

Controlling Antenna Sidelobe Radiation to Mitigate Ku-Band LEO-to-GEO Satellite Interference

ALEX HILLS¹, (Life Fellow, IEEE), JON M. PEHA², (Fellow, IEEE), and JENS MUNK³, (Member, IEEE), SEAN POGORELC⁴

¹Department of Engineering & Public Policy, Carnegie Mellon University, Pittsburgh, PA 15213 USA (e-mail: ahills@cmu.edu)

²Departments of Engineering & Public Policy and ECE, Carnegie Mellon University, Pittsburgh, PA 15213 USA (e-mail: peha@cmu.edu)

³Department of Electrical Engineering, University of Alaska Anchorage, Anchorage, AK 99577 USA (e-mail: jmunk2@alaska.edu)

⁴Department of Electrical and Computer Engineering, Carnegie Mellon University, Pittsburgh, PA 15213 USA (e-mail: spogorel@andrew.cmu.edu)

Corresponding author: Alex Hills (e-mail: ahills@cmu.edu).

This work was supported by the National Science Foundation under Grant 2229067.

ABSTRACT Low-Earth orbiting (LEO) satellites now can provide broadband service anywhere in the world, but they must share the radio spectrum with geosynchronous Earth-orbiting (GEO) satellites. The United Nations International Telecommunications Union (ITU) and the U.S. Federal Communications Commission (FCC) require that operators of LEO satellite systems avoid interfering with GEO satellite systems even though both use the same bands within the radio spectrum. Such interference can occur when a LEO satellite passes through an area between a GEO earth station and its intended GEO satellite. This is called an in-line event. LEO satellites that have beam steering capability can steer their main beams to avoid interfering with GEO earth stations, but even when they do this, LEO satellite antenna side lobes can still cause unacceptable interference. We describe here how these side lobes can be controlled to avoid such interference, and we develop estimates of maximum side lobe levels that must be maintained in currently deployed and future LEO satellite constellations.

INDEX TERMS Communication satellite, in-line event, interference, LEO, NGSO

I. INTRODUCTION

A New group of satellites can deliver worldwide broadband service anywhere on the Earth's surface. They are low Earth orbiting (LEO) satellites, which orbit the Earth at altitudes below 2,000 km. But these new satellites share spectrum with geosynchronous Earth orbiting (GEO) satellites, which orbit at 35,786 km above the equator. This is the altitude at which a GEO satellite revolves around the earth once every 24 hours, and, from a point on the surface of the Earth, appears to stay at a fixed position in the sky. Because they share spectrum, LEO satellite systems can cause interference to GEO satellite systems. In previous work we described the use of beam steering to prevent LEO satellite systems from interfering with GEO satellite systems. In this paper we examine the risk that antenna sidelobe radiation from LEO satellites will cause harmful interference to GEO satellites even when beam steering is used.

Geosynchronous Earth orbit is also called geosynchronous satellite orbit (GSO). But other kinds of satellite orbits are called non-geosynchronous satellite orbits (NGSOs). These

include medium earth orbit (MEO), at altitudes between 2,000 and 35,786 km, and low Earth orbit (LEO), at altitudes below 2,000 km. However, from the perspective of a ground station, NGSO satellites are constantly moving in and out of view, so a single NGSO satellite cannot provide continuous service the way a GSO satellite can. The solution is to deploy a constellation of NGSO satellites so that at least one of them can always cover any individual earth station.

LEO satellites are a class of NGSO satellites that can offer Internet services with low latencies and high throughputs throughout the globe, including regions where the density of likely subscribers is low and where terrestrial infrastructure is unable to support broadband service at a reasonable cost per user. Of the companies that have launched or plan to launch LEO satellites, Starlink and OneWeb are farthest along, having completed most of their first phase constellations with 4,408 and 648 satellites, respectively. Two other companies, Amazon and Telesat plan first phase constellations of 3,236 and 198 LEO satellites to provide worldwide broadband service [1]–[4], and other companies are likely to follow

them.

While the first phase OneWeb and Starlink constellations are now nearly complete, it is expected that the size of these and other constellations will increase dramatically in the future as many more satellites are added. The expanded constellations will include Ku-, Ka-, and V-band satellites.

OneWeb has most of its 648 first phase satellites in orbit. In May 2021 the company filed an application with the U.S. Federal Communications Commission (FCC) to increase the size of its constellation to a total of 47,844 satellites [5]. In January 2021 OneWeb reduced the size of its proposed expanded constellation to 6,372 satellites [6]. But this is still about ten times the number of satellites OneWeb will have in its first phase constellation.

Starlink now has nearly all its 4,408 first phase satellites in orbit. They have permission from the FCC to launch another 12,000 satellites, and, in addition, they have applied for permission to launch an additional 30,000 satellites [7].

Thus, OneWeb and Starlink together now have more than 5,000 satellites in orbit, and they have plans to launch many more. It is likely that other companies will follow them in deploying very large LEO constellations, possibly leading to tens of thousands of LEO satellites in orbit. These very large constellations raise new questions about how the LEO companies will avoid interference to GEO systems [8]. Here we explore the precautions needed for various constellation sizes to avoid causing unacceptable interference and to comply with the regulations of the International Telecommunications Union.

II. LEO-TO-GEO INTERFERENCE

Current and future constellations of LEO satellites share spectrum with GEO satellites in the Ku-, Ka-, and V-bands. A GEO earth station that must receive signals from a GEO satellite at least 35,786 km away cannot tolerate a great deal of interference. Nevertheless, it is the responsibility of the operators of LEO satellites to protect GEO systems from harmful interference. The International Telecommunications Union (ITU), which is the telecommunications arm of the United Nations (UN), has addressed this interference risk in its Radio Regulations for Space Services and requires that LEO satellite operators avoid interfering with GEO systems.

Under ITU regulations, “non-geostationary-satellite systems shall not cause unacceptable interference to ... geostationary satellite networks in the fixed-satellite service and the broadcasting-satellite service” [9]. The operator of an NGSO satellite is free to determine how the satellite will avoid causing harmful interference, which could include adjusting transmit power, adjusting beam direction, or temporarily stopping transmission. These same ITU regulations also state that the operator of an NGSO satellite “shall not claim protection from geostationary satellite networks,” which means that GSO satellites are not expected to alter their design or operation in any way to accommodate NGSO satellites. This is typical of primary-secondary spectrum sharing [10], where in this case the GSO satellites are primary. Many domestic

spectrum regulators, such as the FCC in the United States, have adopted rules that simply incorporate the ITU regulations described above by reference [11]. This means that an NGSO satellite that was licensed by the FCC and that violates these ITU regulations by causing harmful interference to a GSO satellite has also violated U.S. regulations, and the FCC is empowered to enforce those regulations with fines.

A LEO satellite system can interfere with a GEO satellite system during an *in-line event*, an event where a LEO satellite, as it traverses its orbit, passes through an area between the GEO earth station and its intended GEO satellite. Interference can occur when the LEO satellite is either on a line between the GEO earth station and GEO satellite or when it is nearly in line and in an area defined by the beamwidths of the antennas used by the earth stations and satellites [12], [13].

There are two ways that a LEO satellite system can cause interference to a GEO satellite system. First, a LEO satellite transmitting to its intended LEO earth stations (called the “downlink”) may cause interference to a GEO earth station attempting to receive transmissions from its intended GEO satellite. And a LEO earth station transmitting to its intended LEO satellite (called the “uplink”) may cause interference to a GEO satellite attempting to receive transmissions from a GEO earth station. But the distances involved in these two situations are quite different. The distance between a LEO satellite and a LEO earth station is approximately 1,000 km, and the distance between a GEO satellite and a GEO earth station is at least the GEO altitude of 35,786 km. Because the path from a GEO satellite to a LEO earth station is much longer than the path from a LEO satellite to a GEO earth station, LEO satellite to GEO earth station interference is of greater concern and is our focus here.

We have previously described how a LEO satellite can avoid interfering with downlink transmissions from a GEO satellite to a GEO earth station by steering its main beam [14]. In summary, a LEO satellite can avoid downlink interference to GEO earth stations by avoiding pointing its main beam toward any GEO earth station in danger of experiencing objectionable interference. This can happen during an in-line event, which can be represented by passing a line from a GEO satellite to the orbiting LEO satellite, extending this line to the surface of the Earth, and surrounding it by a cone representing the LEO satellite’s main beam. The cone intersects the surface of the Earth in a near ellipse that is this LEO satellite’s forbidden zone that will protect the GEO satellite and its earth stations when the LEO satellite is in this position. If the LEO satellite avoids pointing its beam into this forbidden zone, its main beam will not cause interference to any GEO earth station on the ground using this GEO satellite. We then repeat the process for all visible points on the GEO arc to create an extended forbidden zone, which moves across the Earth’s surface as the LEO satellite traverses its orbit. If the LEO satellite avoids pointing its beam into the extended forbidden zone, its main beam will not cause interference to any GEO earth station [14].

We have shown that the resultant forbidden zones are relatively small and not likely to seriously degrade the performance of a LEO constellation if appropriate beam steering strategies are adopted because a LEO earth station in one LEO satellite's forbidden zone can be served by a different LEO satellite in the same constellation [14]. But off-boresight emissions outside the LEO satellite's main beam resulting from its antenna's side lobes may still cause objectionable interference. And large LEO constellations may be at increased risk of causing objectionable interference to GEO earth stations if the aggregate of their individual satellites' side lobe emissions rises to unacceptable levels. Whereas beam steering assures that the main beam of a LEO satellite does not cause unacceptable interference to a GEO earth station, it does not provide the same assurance for LEO satellites' radiation outside their main beams. Given the large LEO constellations that are planned and proposed, off-boresight radiation outside the main beam can be an interference concern [6].

Here we explore this issue by asking two questions:

- 1) What is the maximum side lobe level that can exist in a Ku-band LEO constellation without causing objectionable interference to a GEO earth station by violating ITU regulations?
- 2) How does this maximum side lobe level change as the size of a LEO constellation increases?

Which LEO satellite off-boresight radiation is of concern depends on the LEO satellites' beam steering strategy. For example, if the LEO satellite avoids pointing its main beam within 10° of any in-line GEO earth station, radiation more than 10° off-boresight is a concern. If, on the other hand, the LEO satellite avoids pointing its main beam within 15° of any in-line GEO earth station, radiation more than 15° off-boresight is a concern.

To illustrate the importance of side lobe levels, we consider LEO-to-GEO interference in the fixed satellite service (FSS) portion of Ku-band. We examine the risk that antenna side lobe radiation will cause objectionable interference to a GEO earth station.

III. INTERNATIONAL TELECOMMUNICATION UNION REQUIREMENTS

Article 22 of the International Telecommunication Union (ITU) Radio Regulations limits the aggregate power flux density that a LEO constellation may cause at a GEO earth station. Since a GEO earth station may be at nearly any location on the Earth's surface, Article 22 puts a limit on the "equivalent power flux density" at any point on the Earth resulting from a LEO constellation's emissions, assuming free space propagation conditions [9].

Article 22 says:

"The equivalent power flux-density, EPFD, at any point on the Earth's surface visible from the geostationary-satellite orbit, produced by emissions from all the space stations of a non-geostationary-satellite system in the

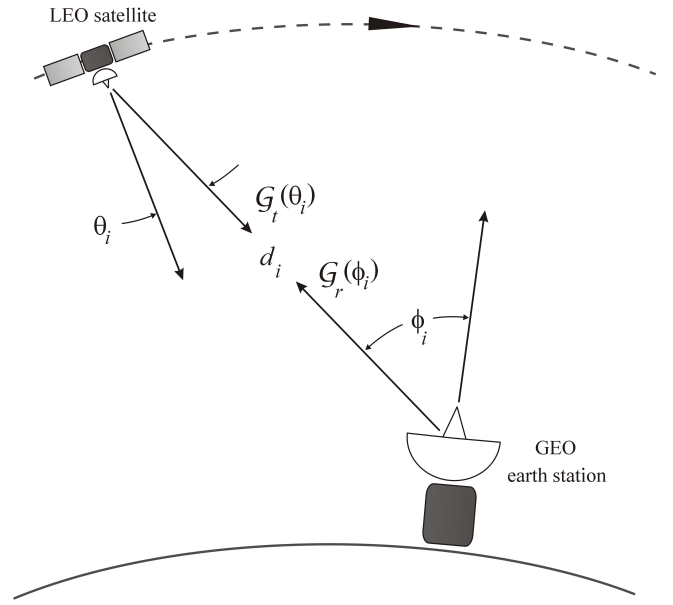


FIGURE 1. LEO satellite's and GEO earth station's off-boresight angles θ_i and ϕ_i in each other's directions and corresponding antenna gains $G_t(\theta_i)$ and $G_r(\phi_i)$

fixed-satellite service ... shall not exceed the limits given [herein]. These limits relate to the equivalent power flux-density which would be obtained under free-space propagation conditions into a reference antenna and in the reference bandwidth ... for all pointing directions towards the geostationary-satellite orbit".

Stated in terms relevant to our questions, the equivalent power flux-density (EPFD) caused by a LEO satellite constellation's emissions must never exceed ITU limits at a reference GEO earth station, which may be located at any point on the Earth's surface.

Power flux density (PFD) is the power density (W/m^2) of radiation arriving at a surface, in this case emissions arriving at the Earth's surface and resulting from the emissions of a constellation of satellites. In dBW/m^2 it is defined as

$$\text{PFD} = 10 \log_{10} \left[\sum_{i=1}^{N_a} \mathcal{P}_i \frac{G_t(\theta_i)}{4\pi d_i^2} \right], \quad (1)$$

where,

- N_a is the number of LEO satellites visible from the reference GEO earth station,
- i is the index of a LEO satellite,
- $\mathcal{P}_i$ is the i^{th} LEO satellite's transmitter power (W),
- θ_i is the i^{th} LEO satellite's off-boresight angle in the direction of a GEO earth station,
- $G_t(\theta_i)$ is the i^{th} LEO satellite's antenna gain (as a ratio) in the direction of the GEO earth station and,

- d_i is the distance (m) between the i^{th} LEO satellite and the GEO earth station.

Note that in equations we use bold face font to denote powers, power flux densities, and gains expressed in dBW, dBW/m², and dB, respectively.

But Article 22 defines “*equivalent power flux density*” (EPFD) by making an adjustment to account for the directivity of a reference GEO earth station antenna. This means that emissions from a LEO satellite contribute to EPFD at reduced levels that depend on the gain of the GEO earth station antenna in the direction of the LEO satellite. For example, emissions from a LEO satellite that is outside the main lobe of a GEO earth station antenna will make a smaller contribution to the EPFD than emissions from a similar satellite that is at boresight, in the center of the GEO earth station’s antenna’s main lobe [9].

Thus, equivalent power flux density EPFD in dBW/m² is defined by Article 22 as

$$\text{EPFD} = 10 \log_{10} \left[\sum_{i=1}^{N_a} \mathcal{P}_i \frac{\mathcal{G}_t(\theta_i) \mathcal{G}_r(\phi_i)}{4\pi d_i^2 \mathcal{G}_{r,max}} \right], \quad (2)$$

where

- ϕ_i is the GEO earth station’s off-boresight angle in the direction of the i^{th} LEO satellite,
- $\mathcal{G}_r(\phi_i)$ is the GEO earth station antenna gain (as a ratio) in the direction of the i^{th} LEO satellite
- $\mathcal{G}_{r,max}$ is the maximum antenna gain (as a ratio) of the GEO earth station antenna.

Fig. 1 illustrates the roles of θ_i , $\mathcal{G}_t(\theta_i)$, ϕ_i , $\mathcal{G}_r(\phi_i)$ and d_i .

Article 22 requires a LEO carrier to limit its satellites’ emissions to assure that the aggregate EPFD arising from all satellites in its constellation never exceeds a specified limit at any point on the Earth’s surface. Article 22 further specifies that each LEO satellite’s transmit power is calculated for a reference bandwidth, which is 40 kHz at Ku-band. The EPFD calculation is then done at Ku-band for reference GEO antennas with diameters of 60 cm, 1.2 m, 3 m, or 10 m [9].

For each GEO earth station antenna size and each possible EPFD level, Article 22 specifies the percentage of time during which various EPFD levels may not be exceeded for each of the four antenna sizes, but Article 22 requires that for fixed satellite service (FSS) at Ku-band an EPFD of -160 dBW/m² may *never* be exceeded [9]. Here we focus on this absolute EPFD limit, which means that for each constellation there is a maximum side lobe level that must never be exceeded because a side lobe above this maximum will at some time be aimed at a point on the Earth’s surface, a potential GEO earth station location.

IV. SIMULATION AND RESULTS

Here we explore the problem of preventing a LEO satellite constellation’s side lobe radiation from exceeding the ITU limit. To approach this, we have created a LEO constellation simulation and used it to find the aggregate EPFD from side lobe radiation at any point on the Earth’s surface. This

Python 3 simulation incorporates the details of any LEO constellation, including its size (number of satellites), orbits, radio frequencies, transmit powers, and antenna patterns, all of which contribute to the constellation’s EPFD at the Earth’s surface. The simulation puts LEO satellites in their orbits with specified effective isotropic radiated powers and side lobe radiation levels, and, for any GEO earth station location, it computes the aggregate EPFD resulting from all LEO satellites visible at that location. For a variety of LEO constellation scenarios, we have used the simulation to create plots of EPFD versus time as LEO satellites traverse their orbits. We use the results to determine the maximum permissible sidelobe level needed to meet ITU requirements for FSS at Ku-band, and we also explore how this maximum sidelobe level changes as a constellation’s size increases.

A. A SINGLE SATELLITE EXAMPLE

For a single LEO satellite, equation (2) expressed in W/m² simplifies to,

$$\text{EPFD}_i = \mathcal{P}_i \frac{\mathcal{G}_t(\theta_i) \mathcal{G}_r(\phi_i)}{4\pi d_i^2 \mathcal{G}_{r,max}} \quad (3)$$

A LEO satellite’s effective isotropic power EIRP (W) is

$$\text{EIRP}_i = \mathcal{P}_i \mathcal{G}_{t,max}, \quad (4)$$

where $\mathcal{G}_{t,max}$ is the maximum gain (as a ratio) of the LEO satellite antenna. Thus, the EPFD (W/m²) of a single LEO satellite is,

$$\text{EPFD}_i = \frac{\text{EIRP}_i \mathcal{G}_t(\theta_i) \mathcal{G}_r(\phi_i)}{4\pi d_i^2 \mathcal{G}_{t,max} \mathcal{G}_{r,max}}, \quad (5)$$

where the second factor is the off-boresight gain (side lobe level) of the LEO satellite transmit antenna in the direction of the GEO earth station, and the third factor is the off-boresight gain of the GEO earth station receive antenna in the direction of the LEO satellite. Each of these can be called the antenna’s relative directivity, its gain as a fraction of its maximum gain.

Equivalently, with EPFD expressed in dBW/m², and EIRP in dBW,

$$\text{EPFD}_i = \text{EIRP}_i - 10 \log_{10} 4\pi - 20 \log_{10} d_i + [\mathbf{G}_t(\theta_i) - \mathbf{G}_{t,max}] + [\mathbf{G}_r(\phi_i) - \mathbf{G}_{r,max}], \quad (6)$$

with

$$\mathbf{G} \equiv 10 \log_{10} \mathcal{G},$$

where the antenna gains $\mathcal{G}_t(\theta_i)$, $\mathcal{G}_{t,max}$, $\mathcal{G}_r(\phi_i)$, and $\mathcal{G}_{r,max}$ are expressed in dB. In equation (6), the fourth term is the LEO satellite transmit antenna gain relative to its boresight gain (relative sidelobe level) in the direction of the GEO earth station, and the fifth term is the GEO earth station receive antenna gain relative to its boresight gain in the direction of the LEO satellite, both expressed in dB.

In the simulation we use equation (6) to calculate the Ku-band EPFD for a single LEO satellite orbiting the Earth at 1,200 km and operating with an EIRP of -3.4 dBW. These are the operating characteristics of a single OneWeb satellite

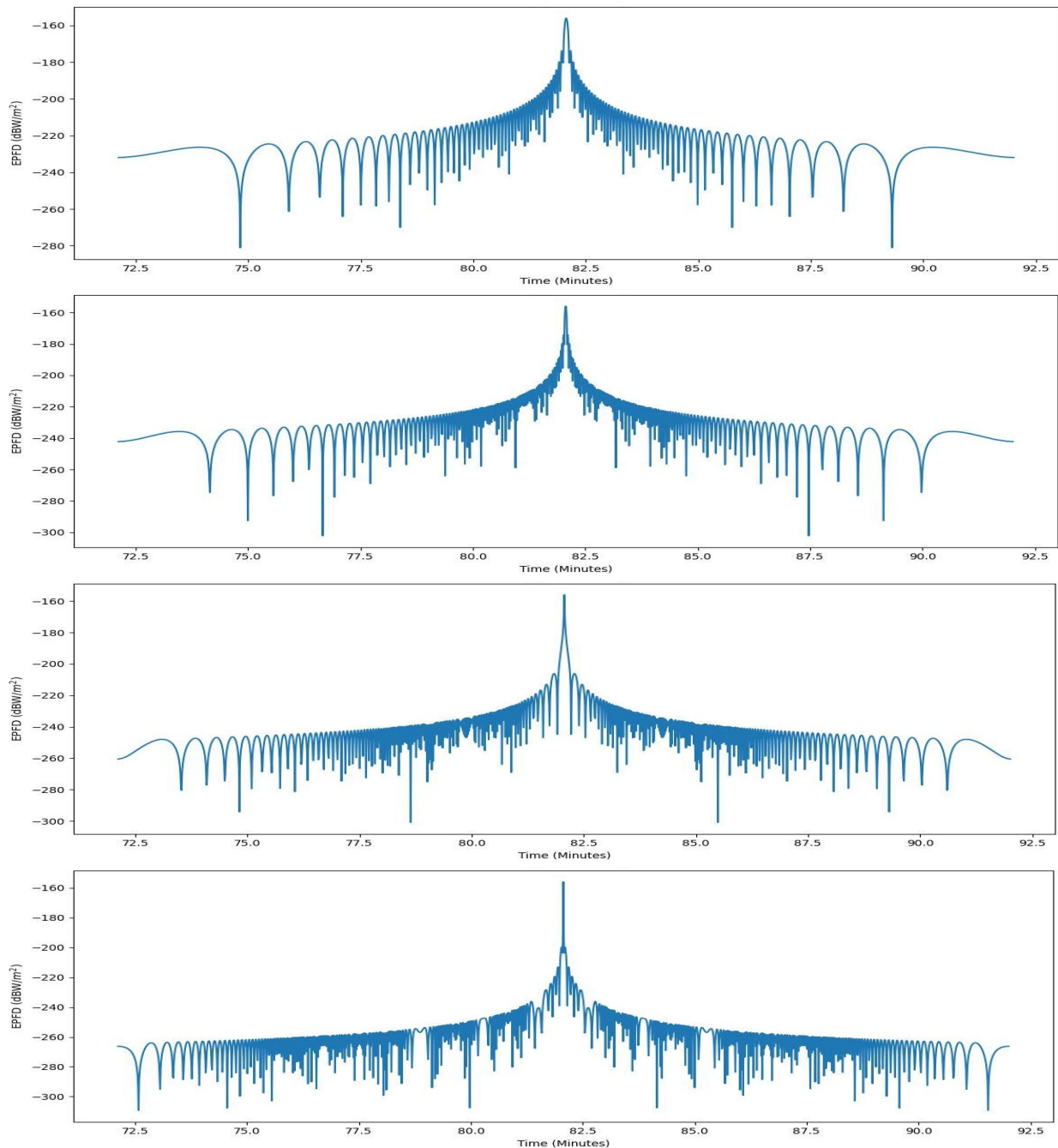


FIGURE 2. EPFD resulting from single OneWeb satellite with -20 dB maximum side lobe vs time for a reference GEO earth station located at the equator with antenna diameter (a) 60 cm, (b) 1.2 m, (c) 3 m, and (d) 10 m

in that company's first phase constellation [15]. We assume a maximum side lobe level of -20 dB and find the resulting EPFD of the radiation that will arrive from this LEO satellite at a 60 cm GEO earth station located at the equator and pointed at a GEO satellite directly overhead. We also assume that the LEO satellite is in an orbit that passes directly above the GEO earth station and therefore will at some moment be in-line with the GEO satellite and earth station, with its highest side lobe directed at the GEO earth station.

Our simulation results, shown in Fig. 2(a), give us the EPFD at the GEO earth station location versus time. The EPFD peaks at -156.0 dBW/m² as the LEO satellite passes directly overhead and in the GEO earth station's main beam, but it then it falls off as the LEO satellite continues along its orbit but is still within the main lobe. Subsequently, the EPFD reaches additional but lower peaks as the LEO satellite passes through the side lobes of the GEO earth station antenna, whose relative directivity is shown in Fig. 3(a). The variations

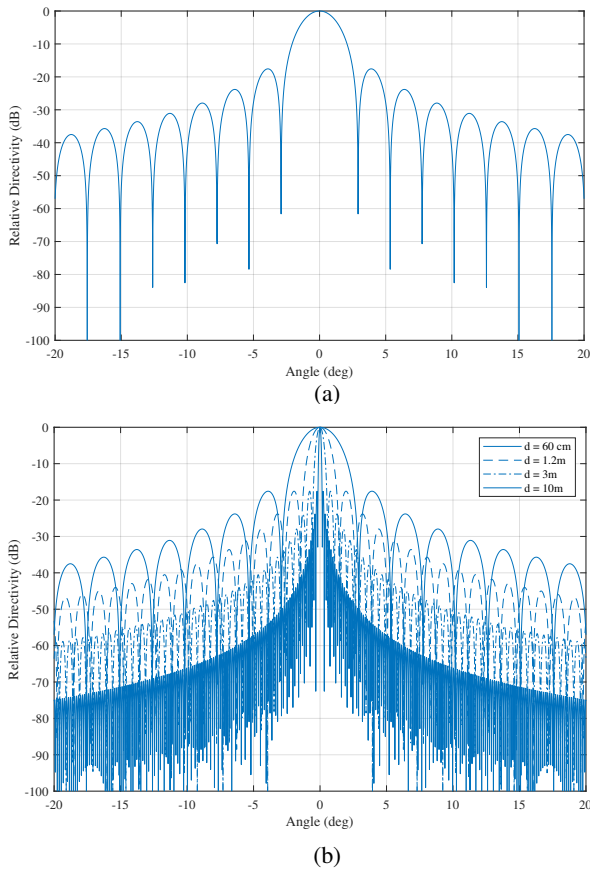


FIGURE 3. Relative directivity (dB) of reference GEO earth station with antenna diameter (a) 60 cm and (b) 60 cm, 1.2 m, 3 m, and 10 m.

in EPFD are explained by the directivity of the GEO earth station antenna and, to a lesser extent, by the increasing path length as the LEO satellite passes overhead and then moves away.

Figs 2(b)-(d) show EPFD for a single LEO satellite and GEO earth station antennas of 1.2, 3 and 10 m diameter, respectively. We note that, whereas all GEO earth station sizes reach EPFD peaks at the same level (-156.0 dBW/m²), the 60 cm antenna results show greater EPFDs over time than the larger earth station antennas because the larger antennas have narrower beams and smaller side lobes, as shown in Fig 3(b). Thus, a GEO earth station with a 60 cm antenna gives the highest EPFD levels over time, and we focus on this antenna size.

Since Article 22 applies everywhere on the Earth's surface, we also consider GEO earth stations at other latitudes but in each case assume that the single LEO satellite passes directly in-line. Peak EPFD results for GEO earth stations at latitudes from 0° to 70° are shown in Fig. 4. The peak EPFD levels are highest for a GEO earth station at the equator because path lengths from an in-line LEO satellite to the GEO earth station are longer for GEO satellites at higher latitudes.

Thus, we find that a GEO earth station with a 60 cm antenna located at the equator and pointed straight up gives

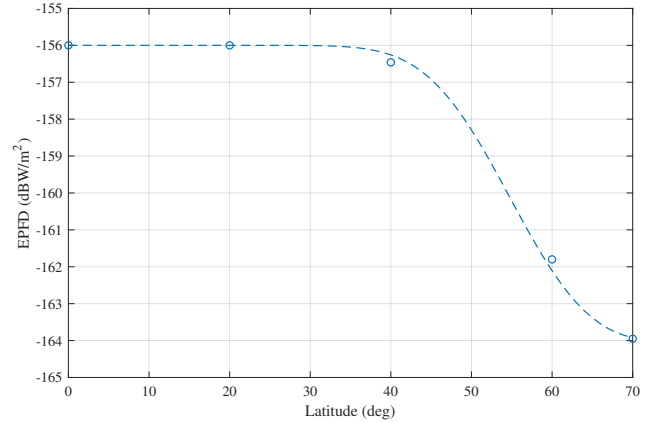


FIGURE 4. Peak EPFD resulting from single OneWeb satellite with -20 dB maximum side lobe vs reference GEO earth station latitude.

the highest peak EPFD, and we continue to focus on this case, shown in Fig. 2(a) for a single satellite with an EIRP of -3.4 dBW orbiting at 1,200 km.

But the peak EPFD received at a GEO earth station from a constellation of LEO satellites with an EIRP of -3.4 dBW may be influenced by the number of LEO satellites that are simultaneously within the GEO earth station's main lobe. With a constellation of many satellites, there may be times when two or more satellites are simultaneously within the GEO earth station's main lobe, and this may increase the peak EPFD to a level higher than that which occurs when only a single LEO satellite is within the main lobe. There may also be some cases where radiation from LEO satellites outside the GEO earth station's main lobe contributes substantially to the aggregate EPFD. We consider these possibilities by simulating two real LEO constellations, those of OneWeb and Starlink.

We simulated these two constellations to find acceptable side lobe levels for each in the context of the ITU requirement. Starlink uses orbital altitudes between 530 and 570 km, and OneWeb uses an orbital altitude of 1,200 km. These orbital altitudes are near the low and high ends of the range of altitudes expected to be used by other LEO constellations [1], [2].

B. ONEWEB

The first phase OneWeb constellation has 12 orbital planes with 49 operational satellites per plane, a total of 588 operational satellites. OneWeb also plans an additional 60 in-orbit spare satellites for a total of 648 satellites, but these non-operational spares were not part of our simulation. The OneWeb satellites orbit at 1,200 km with a 40 kHz EIRP of -3.4 dBW, the same altitude and EIRP we used in our single satellite simulation [15].

In Fig. 5(a) we show a plot of EPFD versus time for the first phase OneWeb constellation. Again, we show the EPFD for a 60 cm GEO earth station located at the equator.

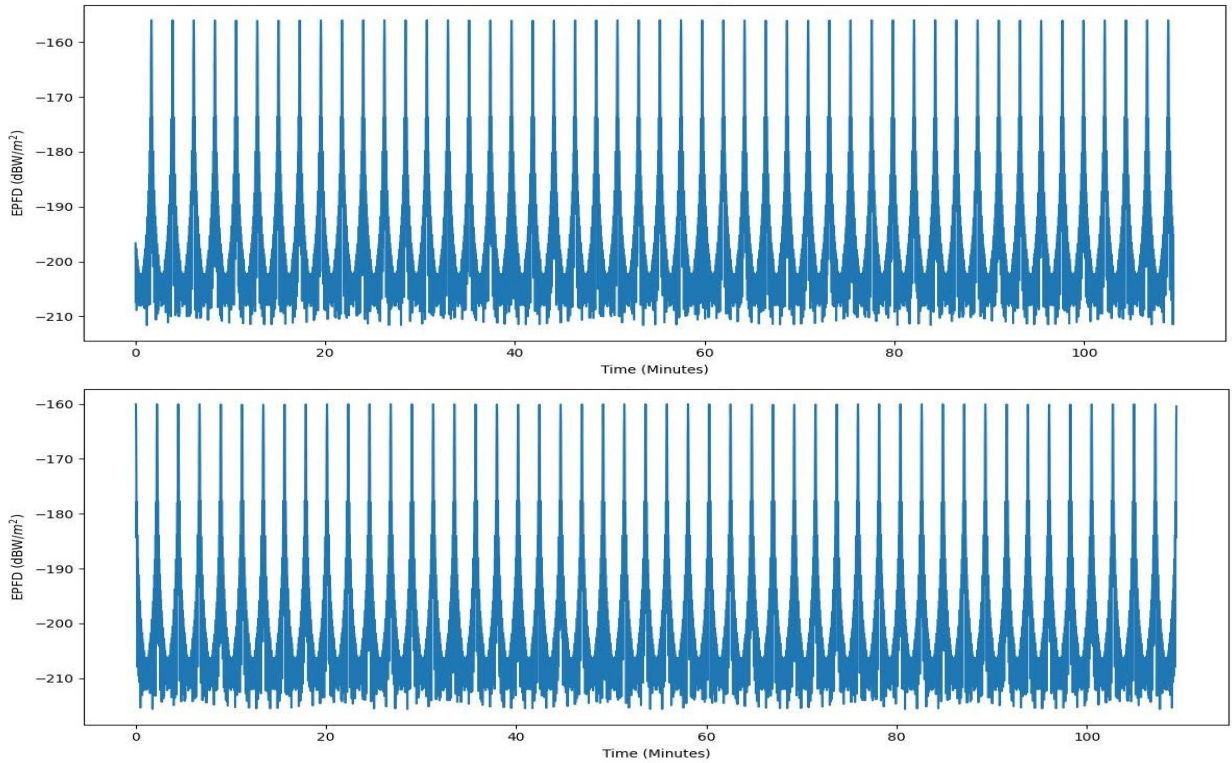


FIGURE 5. EPFD resulting from current OneWeb constellation vs time for (a) -20.0 dB maximum side lobe and (b) -24.0 dB maximum side lobe

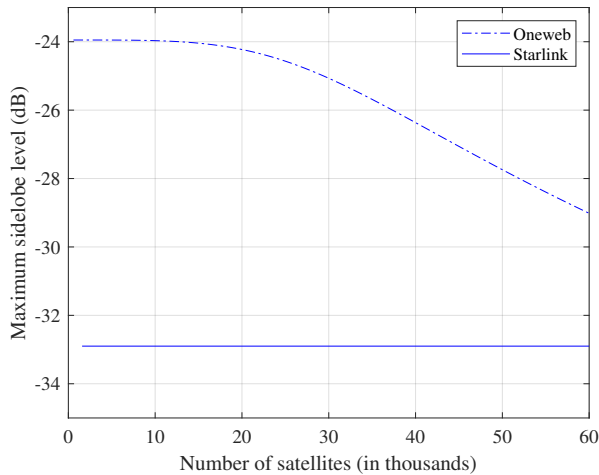


FIGURE 6. Maximum permissible side lobe level (dB) for OneWeb and Starlink Group 1 Ku-band constellations vs number of satellites in each constellation

We see that the EPFD peaks every 2.23 minutes, as each satellite passes through the main lobe of the GEO earth station antenna. The EPFD reaches a peak value of -156.0 dBW/m^2 , the same level as the peak EPFD shown in Fig. 2(a) for a single satellite passing through the main lobe. In the 588-satellite constellation, there is never more than one satellite in the GEO earth station's main lobe, and only this satellite makes a significant contribution to EPFD.

But this -156.0 dBW/m^2 peak value occurs with a maximum side lobe level of -20 dB and exceeds the ITU limit. To limit the EPFD to the -160 dBW/m^2 , we reduce the maximum LEO satellite side lobe level to -24.0 dB . With this done, the peak EPFD is exactly -160 dBW/m^2 , as shown in Fig. 5(b). Thus, for the first phase OneWeb constellation, the maximum side lobe level must be -24.0 dB to meet the ITU requirement. This is true when the LEO main beam is steered appropriately using either “progressive pitch,” as has been described by OneWeb [16] or by using an electronically steerable antenna.

But as OneWeb increases the number of satellites in its constellation, multiple satellites can appear in the reference GEO earth station's main beam and be close enough to each other to cause the aggregate EPFD to rise above the ITU limit. At this point it will be necessary to reduce the maximum side lobe level. We have used simulation to determine the OneWeb constellation size at which this occurs (15,000 satellites) and find the maximum side lobe level that will meet the ITU Article 22 requirement for OneWeb constellations ranging from 588 satellites to 60,000 satellites. The results are shown in Fig. 6. As the constellation size increases to 60,000 satellites, the peak EPFD increases to -155.9 dBW/m^2 , requiring that the maximum side lobe level be reduced from -24.0 dB to -28.9 dB .

In our simulation we have assumed that the satellites in each orbital plane are uniformly spaced. With large constellations, such uniform spacing reduces the size of the side lobe level reduction needed to stay within the ITU limit.

C. STARLINK

Starlink's first phase constellation of 4,408 satellites includes 1,584 Group 1 satellites, which are in an orbit at 550 km altitude with 53.0° inclination and have user links at Ku-band. Group 1 has 72 orbital planes with 22 satellites per plane and Ku-band user links with a 40 kHz EIRP of -1.3 dBW. Starlink's remaining first phase satellites link to users at Ka-band [1], [2], [15].

Fig. 7(a) shows simulated EPFD versus time for these first phase Ku-band satellites. Again, we show the EPFD for a 60 cm GEO earth station located at the equator. But because the Starlink satellites are at a lower altitude than OneWeb and closer to the reference GEO earth station, we anticipate that a lower side lobe level will be needed and assume a maximum side lobe level of -30 dB. We see that the EPFD peaks every 4.34 minutes, as each Starlink satellite passes through the main lobe of the GEO earth station antenna. Like the first phase OneWeb constellation, Starlink's 1,584-satellite Ku-band constellation has only one satellite at a time in the main lobe. The EPFD reaches a peak value of -157.1 dBW/m², the same level as the peak EPFD for a single satellite passing through the main lobe. Only the satellite in the main lobe makes a significant contribution to EPFD.

But this -157.1 dBW/m² peak value occurs with a maximum side lobe level of -30 dB and exceeds the ITU limit. To limit the EPFD to -160 dBW/m², we reduce the maximum LEO side lobe level to -32.9 dB. With this done, the peak EPFD is exactly -160 dBW/m², as shown in Fig. 7(b). Thus, for the first phase Ku-band Starlink constellation, the maximum side lobe level must be -32.9 dB to meet the ITU requirement even when the LEO main beam is steered appropriately.

We have also used simulation to find the maximum side-lobe level for increasing numbers of satellites in the Starlink constellation to determine if the resultant peak EPFD increases. But because there are only 22 satellites in each plane, even large constellations up to 60,000 satellites do not cause the EPFD to rise above -160 dBW/m², and the maximum sidelobe level remains at -32.9 dB. As shown in Fig 6, a maximum side lobe level of -32.9 dB will meet the ITU Article 22 requirement for similar 72 plane Starlink Ku-band constellations ranging from 1,584 satellites up to 60,000 satellites.

V. CONCLUSIONS

We return to the two questions we set out to address:

- 1) What is the maximum side lobe level that can exist in a Ku-band LEO constellation without causing objectionable interference to a GEO earth station by violating ITU regulations?
- 2) How does this maximum side lobe level change as the size of a LEO constellation increases?

For small to mid-size constellations, the maximum side lobe level for a constellation can be found by starting with a single satellite passing within the main lobe of a reference

60 cm GEO earth station antenna. This maximum side lobe level depends on the EIRP of each satellite and the altitude of the constellation.

For example, a single OneWeb satellite in the current constellation, orbiting at 1,200 km with an EIRP of -3.4 dBW, must maintain a maximum side lobe level of -24.0 dB to comply with the ITU Article 22 Ku-band fixed satellite service (FSS) EPFD requirement. But the current Starlink Group 1 satellites orbit at 550 km with an EIRP of -1.3 dBW, and a single satellite must maintain a maximum side lobe level of -32.9 dB to comply with the ITU requirement. Starlink's lower maximum side lobe level is primarily the result of its lower altitude but also results from its slightly higher EIRP.

We have seen that a single satellite's maximum EPFD occurs at the GEO earth station's boresight when the LEO satellite is directly overhead. To keep this maximum EPFD at or below -160 dBW/m², based on equation (6), the LEO satellite's side lobes must be controlled to assure that

$$G_t(\theta_i) - G_{t,max} \leq -160 \text{ dBW/m}^2 - \text{EIRP}_i \quad (7) \\ + 10 \log_{10} 4\pi + 20 \log_{10} d_i,$$

where $G_t(\theta_i) - G_{t,max}$, is the LEO satellite's sidelobe level, and d_i is at its minimum value, which is the orbital altitude.

Thus, if OneWeb or Starlink changes its Ku-band satellite characteristics, or if another carrier plans a new Ku-band constellation, the company can use equation (7) to find the single satellite maximum side lobe level needed to comply with ITU requirements, using only the satellite's EIRP and orbital altitude.

As shown in Fig. 6, this single satellite calculation is valid for an entire constellation up to a threshold constellation size. For constellations whose size is above the threshold, the maximum side lobe level will be modestly reduced for some cases where multiple LEO satellites simultaneously pass within the main beam of the GEO earth station. The amount of this reduction depends on the details of the constellation's satellite orbits and can be determined by examining such cases, which are less likely to occur if satellites are uniformly spaced within each orbital plane.

In summary, compliance with Article 22 of the ITU Radio Regulations requires that LEO satellite systems use proper beam steering of each LEO satellite's main beam and shape the LEO satellite's antenna pattern so that radiation outside the main beam is appropriately controlled. This means that the satellite antenna's side lobes must not rise above a maximum level. This level depends on the satellite's EIRP, its orbital altitude, the number of satellites in each orbital plane, and the spacing of satellites within each plane.

Article 22 applies to a single LEO satellite system. It does not apply to multiple LEO satellite systems sharing the same spectrum with each other and, of course, with GEO satellite systems. But one can easily imagine a situation in which each of two LEO satellite systems are in compliance with the EPFD limit but the aggregate EPFD of the two systems

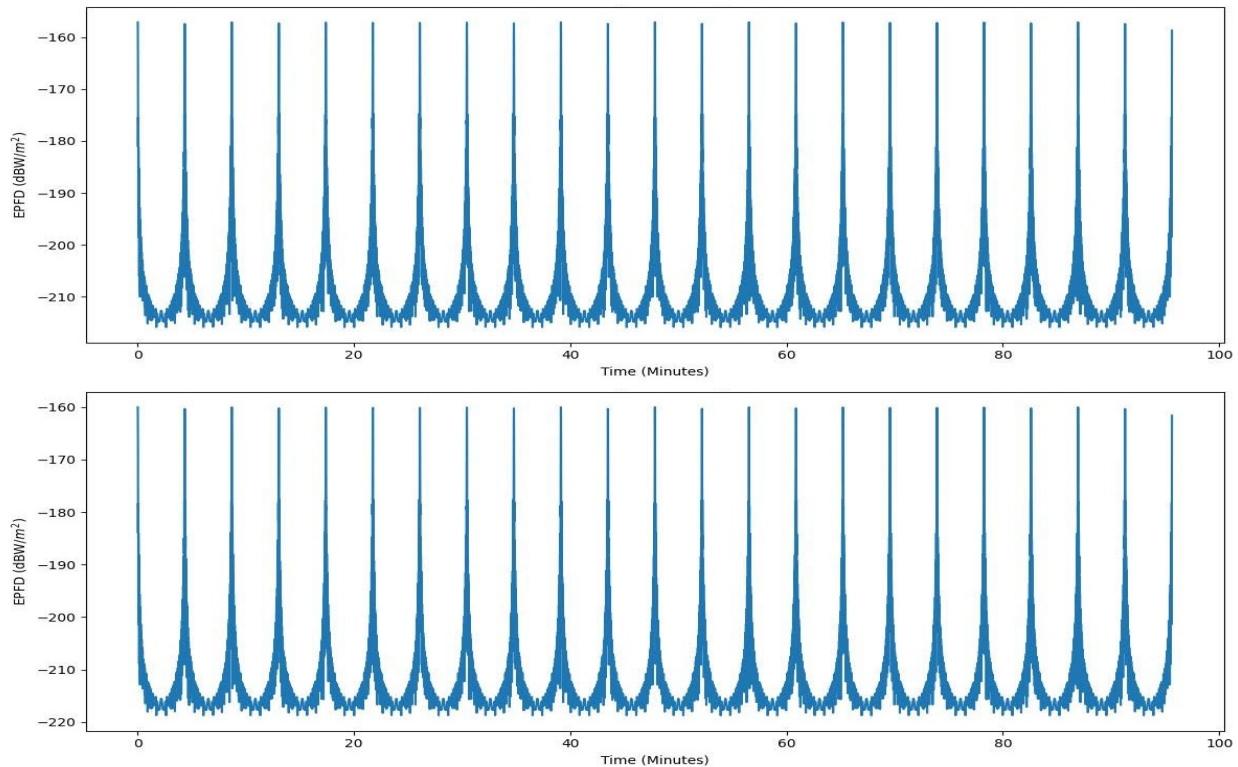


FIGURE 7. EPFD resulting from current Starlink Group 1 Ku-band constellation versus time for (a) -30.0 dB maximum side lobe and (b) -32.9 dB maximum side lobe

exceeds the limit. And with three or more LEO systems sharing the same spectrum, the situation could be even worse. For this reason, we suggest that the ITU consider revising Article 22 so that it applies to two or more LEO systems operating in the same spectrum.

REFERENCES

- [1] I. del Portillo, B. G. Cameron, and E. F. Crawley, "A Technical Comparison of Three Low Earth Orbit Satellite Constellation Systems to Provide Global Broadband", *Acta Astronautica*, 2019.
- [2] N. Pachler, I. del Portillo, E. F. Crawley, and B. G. Cameron, "An Updated Comparison of Four Low Earth Orbit Satellite Constellation Systems to Provide Global Broadband", *2021 IEEE International Conference on Communications Workshops (ICC Workshops)*, 2021, pp. 1-7, doi: 10.1109/ICCWorkshops50388.2021.9473799.
- [3] A. Jones, "Amazon Gets a Green Light to Launch 3000-Satellite Kuiper Constellation," *Space.com*, February 23, 2023, <http://space.com>
- [4] J. Rainbow, "Telesat to Order 100 Fewer Satellites for LEO Constellation," *Space News*, May 6, 2022, <https://spacenews.com>
- [5] WorldVu Satellites, Ltd., "Application for Modification: Modification to OneWeb U.S. Market Access Grant for the OneWeb Ku- and Ka-band System," filed with the Federal Communications Commission, May 26, 2020.
- [6] J. Foust, "OneWeb Slashes Size of Future Satellite Constellation," *Space News*, January 14, 2021, <https://space news.com>
- [7] M. Wall, "SpaceX Launches 56 Starlink Satellites, Lands Rocket at Sea," *space.com*, March 29, 2023, <http://space.com>
- [8] ViaSat, Inc., "Ensuring Competition and Innovation in the New Space Age," ViaSat, Inc., Carlsbad, California, February 2023.
- [9] International Telecommunications Union, ITU-R (2016). Radio Regulations. Chapter VI, Provision of services and stations. Article 22, Space Services ("ITU-R Article 22"). Available: <https://life.itu.int/radioclub/rr/art22.pdf>
- [10] J. M. Peha, "Sharing Spectrum Through Spectrum Policy Reform and Cognitive Radio," in *Proceedings of the IEEE*, vol. 97, no. 4, pp. 708-719, April 2009, doi: 10.1109/JPROC.2009.2013033.
- [11] Code of Federal Regulations, "Protection of GSO networks by NGSO systems", 47 CFR §25.289.
- [12] H. Al-Hraishawi, S. Chatzinotas and B. Ottersten, "Broadband Non-Geostationary Satellite Communication Systems: Research Challenges and Key Opportunities", *2021 IEEE International Conference on Communications Workshops (ICC Workshops)*, 2021, pp. 1-6, doi: 10.1109/ICCWorkshops50388.2021.9473786.
- [13] C. Braun, A. M. Voicu, L. Simić and P. Mähönen, "Should We Worry About Interference in Emerging Dense NGSO Satellite Constellations?", *2019 IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN)*, 2019, pp. 1-10, doi: 10.1109/DySPAN.2019.8935875.
- [14] A. Hills, J.M. Peha, and J. Munk, "Feasibility of Using Beam Steering to Mitigate Ku-band LEO to GEO Interference", *IEEE Access*, pp. 74023-74,032, July 2022.
- [15] S. Xia, Q. Jiang, C. Zou, and G. Li, "11Beam Coverage Comparison of LEO Satellite Systems Based on User Diversification," *IEEE Access*, pp. 181656-181,667, December 2019.
- [16] T. Li, J. Jin, W. Li, Z. Ren, and L. Kuang, "Research on Interference Avoidance Effect of OneWeb Satellite Constellation's Progressive Pitch Strategy," *Int. J. Satell. Commun. Netw.*, vol. 35, no. 9, pp. 524-538, 2021, doi: 10.1002/SAT.1339



ALEX HILLS (LF'08) received a B.S. in electrical engineering from Rensselaer Polytechnic Institute, an M.S. in electrical engineering from Arizona State University, and a Ph.D. in engineering and public policy from Carnegie Mellon University. He is Distinguished Service Professor in the Department of Engineering and Public Policy at Carnegie Mellon, where he has also served as the university's Vice Provost and Chief Information Officer and as the first Director of its Information

Networking Institute.

In 1993 Dr. Hills founded and led Carnegie Mellon's Wireless Andrew project, a pre-802.11 campus-wide high-speed wireless network that became a model for later large Wi-Fi networks. Before he joined Carnegie Mellon, Professor Hills lived and worked in Alaska, where during the 1970s he was involved in pioneering the use of small satellite earth stations to serve the state's many villages. He has served as Alaska's Deputy Commissioner of Administration and chief telecommunications official and as a U.S. Army officer and company commander in South Korea.

Professor Hills has held visiting appointments as Profesor Extraordinario at the Universidad Austral de Chile in Valdivia, Chile, as Cheng Tsang Man Professor at Nanyang Technological University in Singapore, and as Erskine Fellow at the University of Canterbury in Christchurch, New Zealand. An inventor with 17 patents, he is a member of the Alaska Innovators Hall of Fame. In 2007 he was named the IEEE Region 6 Engineer of the Year and the same year the state's engineering societies named him Alaska's Engineer of the Year. In 2014 he was awarded an honorary doctorate by the University of Alaska Anchorage.



JON M. PEHA (F'12) received a B.S. from Brown University and a Ph.D. in electrical engineering from Stanford University. He is a Professor at Carnegie Mellon University in the Department of Engineering & Public Policy and the Department of Electrical & Computer Engineering, Director of CMU's Center for Executive Education on Technology Policy, and former Associate Director of the Center for Wireless and Broadband Networks.

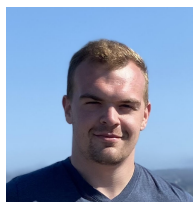
Dr. Peha has served at the Federal Communications Commission (FCC) as Chief Technologist, in the White House as Assistant Director of the Office of Science and Technology Policy, in industry as Chief Technical Officer of three high-tech companies, on the staff of the House of Representatives Energy and Commerce Committee, and at USAID, where he helped launch and then led a U.S. government interagency program to assist developing countries with information infrastructure. His research spans technical and policy issues of information and communications technologies.

He has received the FCC's Excellence in Engineering Award for his contributions to the U.S. National Broadband Plan, the IEEE Communications Society's TCCN Publication Award for career contributions to spectrum management, and the Brown University Engineering Medal. He is a Fellow of the AAAS and was selected as one of 40 Featured AAAS Science and Technology Policy Fellows of the last 40 years ("40@40").



JENS MUNK (M'97) was born in Copenhagen, Denmark in 1960. He received the B.S. and the M.S. in electrical engineering, from The Ohio State University, Columbus, Ohio in 1990 and 1992, respectively. He received the Ph.D. in geophysics in 1999, again from Ohio State. From 1999 to 2001 he was a Post-Doctoral Researcher at the Byrd Polar and Climate Research Center - The Ohio State University.

Since 2001 Dr. Munk has been with the University of Alaska Anchorage and currently holds the title Professor and Chair with the Electrical Engineering Department. His research interests include inverse problems, numerical methods in electromagnetics, remote-sensing of ice-sheets, and the application of shallow geophysical methods to better understand the impact of climate change in Arctic environments.



SEAN POGORELC received the B.S. degree in Electrical and Computer Engineering from Carnegie Mellon University in spring 2021 and the M.S. degree in Electrical and Computer Engineering from CMU in spring of 2023. Working as a research assistant, he participated in this LEO-to-GEO satellite interference work during the 2022-23 academic year. While a Carnegie Mellon student, he was a football player, captain of the varsity football team, a repeat Academic All-

Conference recipient, and an All-American Offensive Lineman. He was a graduate assistant football coach for the Carnegie Mellon team from spring 2022 until spring 2023.

...